

Q1. Exception

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.*;

class Main{
    public static void main(String[] args){
        try{
            Scanner s = new Scanner(System.in);
            int n = s.nextInt();
            int[] a = new int[n];
            for(int i=0;i<5;i++){
                a[i] = s.nextInt();
            }
            int k = s.nextInt();
            System.out.println(a[k]);
        }
        catch(Exception ae){
            System.out.println("Exception: Invalid array index.");
        }
    }
}
```

INPUT

```
6
10 20 30 40 50
```

OUTPUT

Exception: Invalid array index.

Q2. IllegalArgumentException

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.*;

class Main{
    public static void main(String[] args){
        Scanner s = new Scanner(System.in);
        try{
            int marks = s.nextInt();
            if(marks<0||marks>100)
                throw new IllegalArgumentException();
            System.out.println(marks);
        }
        catch(IllegalArgumentException e){
            System.out.println("Exception: Invalid marks. Marks should be between 0 and 100.");
        }
    }
}
```

INPUT

120

OUTPUT

Exception: Invalid marks. Marks should be between 0 and 100.

Q3. user-defined exception named WeakPasswordException.

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.*;

class Main{
    public static void main(String[] args){
        Scanner s = new Scanner(System.in);
        String pass = s.next();
        try{
            if(pass.length()<8)
                throw new WeakPasswordException("Password must contain at least 8 characters.");
            System.out.println("Password accepted.");
        }
        catch(WeakPasswordException e){
            System.out.println("Exception: "+e.getMessage());
        }
    }
}

class WeakPasswordException extends Exception{
    WeakPasswordException(String s){
        super(s);
    }
}
```

INPUT

java 123

OUTPUT

Exception: Password must contain at least 8 characters.

Q4. Email Validation

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.*;

class Main{
    public static void main(String[] args){
        Scanner s = new Scanner(System.in);
        String email = s.next();
        try{
            if(!email.contains("@"))
                throw new InvalidEmailException("Invalid email address.");
            System.out.println("Valid email address.");
        }
        catch(InvalidEmailException e){
            System.out.println("Exception: "+e.getMessage());
        }
    }
}

class InvalidEmailException extends Exception{
    InvalidEmailException(String s){
        super(s);
    }
}
```

INPUT

student,gmail.com

OUTPUT

Exception: Invalid email address.

Q5. Banking Transaction Using Exception Propagation

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        sc.next();
        int balance = sc.nextInt();

        sc.next();
        int withdrawAmount = sc.nextInt();

        try {
            withdraw(balance, withdrawAmount);
        } catch (InsufficientBalanceException e) {
            System.out.println("Exception propagated to main().");
            System.out.println("InsufficientBalanceException: " + e.getMessage());
        }
    }

    static void withdraw(int balance, int withdrawAmount) throws InsufficientBalanceException {
        validateBalance(balance, withdrawAmount);
    }

    static void validateBalance(int balance, int withdrawAmount) throws InsufficientBalanceException {
        processTransaction(balance, withdrawAmount);
    }

    static void processTransaction(int balance, int withdrawAmount) throws InsufficientBalanceException {
        if (withdrawAmount > balance) {
            throw new InsufficientBalanceException("Withdrawal amount exceeds available balance.");
        }
    }
}

class InsufficientBalanceException extends Exception {
    InsufficientBalanceException(String message) {
        super(message);
    }
}
```

INPUT

Balance:
5000
Withdraw:
7000

OUTPUT

Exception propagated to main().
InsufficientBalanceException: Withdrawal amount exceeds available balance.